

SUMMARY

Data Science Engineer with 3+ years' experience uniting data engineering and machine learning to design pipelines, build predictive models, and drive data-driven decision making.

WORK EXPERIENCE**Allwyn Corporation** | AI Engineer

Jan 2025 – May 2025

- Developed a RAG-based enterprise chatbot to streamline proposal generation using LLMs and AWS.
- Enhanced automation of past performance extraction for government contracting proposals, reducing manual effort by over 60%.
- Engineered and deployed a production-grade RAG pipeline using Amazon QnABot, integrating Amazon Lex, Bedrock (Claude 3.5, Jamba, Llama 3), Lambda, Kendra, and S3, cutting down proposal search time by approximately 65%.
- Indexed over 1,500+ multi-format documents from SharePoint into Amazon Kendra, achieving more than 92% precision in relevant passage retrieval and reducing document miss rate by 40%.
- Implemented a robust evaluation framework using Python, Pandas, and Streamlit to assess 31 real-world queries across 3 LLMs, improving output relevance and formatting by more than 50% through prompt versioning.

Bank of America | Technical Analyst

Sep 2021 – Jun 2023

- Developed and productionized machine learning models (XGBoost, Logistic Regression) to detect fraudulent credit card transactions, improving recall (fraud capture rate) by 21% while reducing missed fraud cases by 28%.
- Engineered domain-specific features from streaming transaction data, including velocity checks, merchant risk profiling, and geolocation anomaly detection.
- Designed a fraud scoring framework that combined model predictions with business rules, enabling real-time prioritization of high-risk activities across millions of daily transactions.
- Applied time-series analysis on transaction sequences to identify abnormal spending behaviors and synthetic identity patterns, contributing to early detection of organized fraud rings.

Apollo Hospitals | Data Analyst Intern

Jan 2021 – Jul 2021

- Automated ETL pipelines for patient and operational data using SQL, Python (Pandas, NumPy), and Apache Airflow, cutting manual Excel-based data prep and reporting by ~60%.
- Built Tableau dashboards to track KPIs (bed occupancy, ER turnaround, discharge times) with live data refresh, enabling department heads to reallocate staff and resources more efficiently (approx. 30% faster response to surges).
- Maintained predictive models (Scikit-learn, XGBoost) to forecast patient admission volumes and seasonal spikes, supporting staffing and supply chain planning.
- Unified disparate sources (EMR, billing, lab data) into Snowflake via SQL transformations and data lake staging, improving cross-department reporting consistency.

EDUCATION**Master of Science in Data Analytics Engineering**

Aug 2023 - May 2025

George Mason University, Virginia | GPA 3.9/4.0

Bachelor of Engineering in Computer Science and Engineering

Jun 2017 - May 2021

SRM Institute of Science and Technology | GPA 3.6/4.0

PROJECTS**YouTube Trending Analytics Pipeline** | Technologies: Python | Youtube API | Pandas | Amazon SageMaker | AWS S3 | Looker Studio | BigQuery

- Created an end-to-end cloud pipeline to fetch trending videos via the YouTube Data API (Python on SageMaker), transform to Parquet, land in S3, and model in BigQuery using a STAR schema for fast slicing by region, category, and channel. Delivered an interactive Looker Studio report (Daily Summary and category/country breakdowns) with owner-credentialed data sources for public sharing; designed the roadmap for automated refresh, a Historical Trends page, and AI-generated daily summaries with a read-only data chat.

AWS-Powered COVID-19 Data Lake Analytics Pipeline | Technologies: AWS -S3, Glue, Lambda, Athena, QuickSight, CloudWatch | Python | SQL

- Implemented a serverless data pipeline using AWS S3, Glue, and Lambda to automate the ingestion, transformation, and storage of raw CSV files, enabling fully managed ETL workflows. Queried curated datasets via Athena and created QuickSight dashboards to deliver actionable business insights, reducing manual reporting efforts by 80%. Orchestrated real-time data processing and monitoring using CloudWatch, achieving a scalable, production-ready pipeline aligned with modern data engineering practices.

TECHNICAL SKILLS

- **Generative AI & ML:** Predictive Modelling | Regression | A/B Testing | LLM | RAG | NLP | Vector databases | Prompt Engineering
- **Data Analysis & Visualization:** Python | Pandas | NumPy | Advanced Excel | Tableau | Power BI | Matplotlib | Seaborn | Plotly | Looker
- **Big Data & Cloud :** Google Cloud | AWS | Snowflake | Apache Spark | Kafka | HDFS | Databricks
- **Data Engineering & Pipelines:** Apache Airflow | ETL | PySpark | SQL Alchemy
- **Programming & Databases:** Python | R | SQL | PL/SQL | Oracle | Postgres | BigQuery | S3 | MySQL | SSIS.
- **DevOps & CI/CD :** Docker | Kubernetes | Jenkins | Git